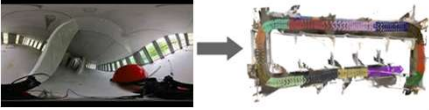


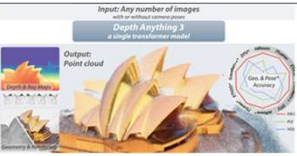
1. INTRODUCTION:

Goal: Generate accurate 3D reconstructions of indoor construction sites from sparse and noisy 360° walkthrough videos.

**Challenges:**

- Repetitive Structures
- Motion Blur & Low Light
- Long Trajectories
- Dynamic Objects

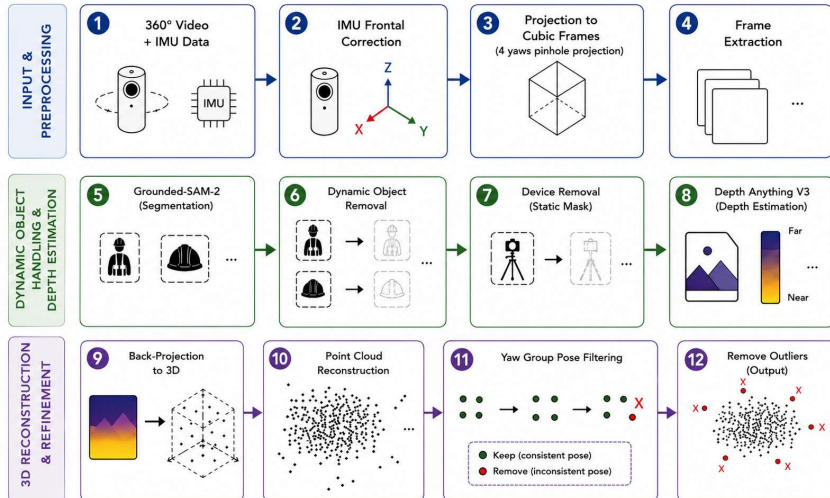
2. MODEL ARCHITECTURE:

**Depth Anything 3**

- Feed-forward transformer for 3D scene reconstruction
- Accepts images with or without known camera poses
- Jointly predicts dense depth maps and camera poses
- Produces dense 3D point clouds

3. METHODOLOGY:

Our method was tested on the **Hilti x Trimble Challenge 2026 Dataset**, captured directly on active construction sites using 360° camera with IMU.



4. RESULTS:

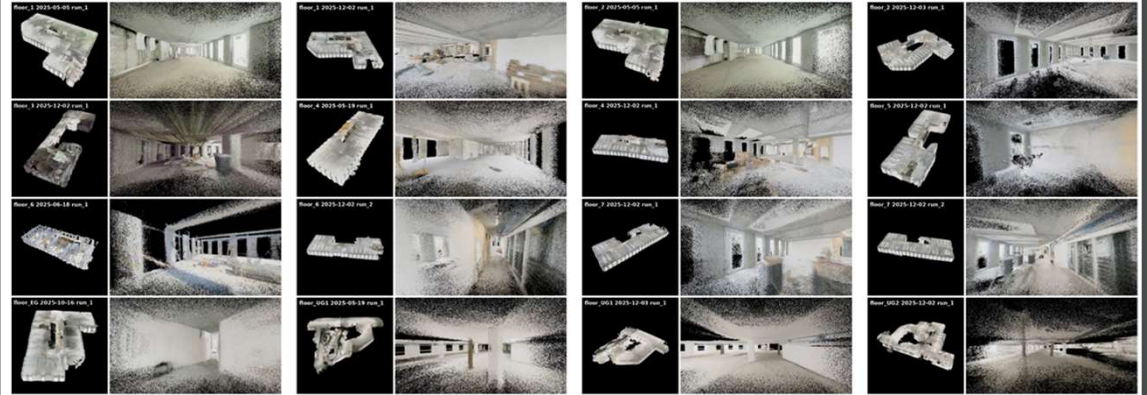
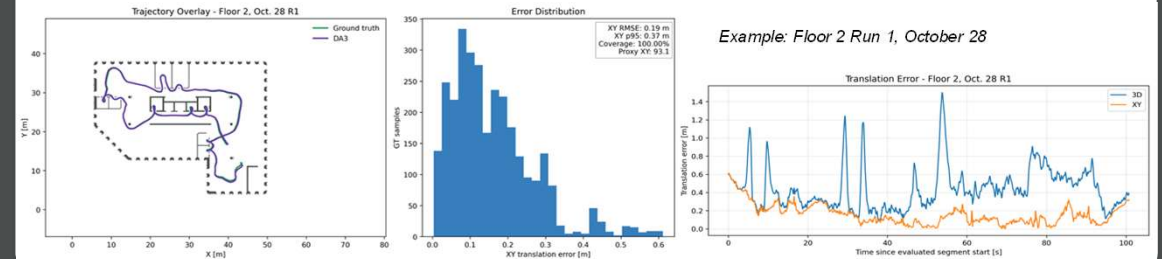
Qualitative Results**Quantitative Results**

Table 1: Ground-truth trajectory evaluation.

Run	Cov. (%)	XY error		3D error		Proxy XY	1s RPE	Sim(3) scale
		RMSE	p95	RMSE	p95			
Floor 1, May 5	100.0	0.314	0.605	0.734	0.936	88.8	0.074	1.001
Floor 2, May 5	100.0	1.027	1.651	1.344	1.932	67.0	0.119	0.956
Floor 2, Oct. 28 R1	100.0	0.193	0.366	0.495	0.873	93.1	0.276	0.990
Floor 2, Oct. 28 R2	100.0	0.579	1.080	1.140	2.159	78.6	0.892	0.986
Floor UG1, Oct. 16	100.0	3.261	5.660	3.372	5.728	34.8	0.249	0.917

Table 2: Mean evaluation results over all ground-truth runs.

Metric	Mean value
Map XY RMSE (m)	1.075
Map XY p95 (m)	1.872
Proxy XY score	72.5
Map 3D RMSE (m)	1.417
Sim(3) 3D RMSE (m)	0.725
Mean Sim(3) scale	0.970



5. CONTRIBUTIONS AND LIMITATIONS:

Contributions

- Panorama support via four-yaw pinhole projections
- Stable dense reconstruction on most challenge sequences
- Dynamic object removal
- Improved loop closure and trajectory correction
- Camera pose filtering and outlier cleaning

Limitations

- Undetected pose outliers remain → fake structures
- Reflective surfaces → missing geometry
- Complex motion / clutter → reconstruction artifacts
- Over-/under-exposure → unstable depth estimation
- Conservative loop detection → missed loop closures